POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2025
Programme: BE | | Full Marks : 100
Course: Calculus II | | Pass Marks : 45
| | Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1.

a) Evaluate $\int_0^2 \int_0^{\sqrt{4-y^2}} \cos(x^2+y^2)dxdy$ by changing into polar integration. 5

b) Evaluate: $\int_0^1 \int_0^1 \int_0^1 (x^2+y^2+z^2)dzdydx$ 5

c) Find the volume in the first octant bounded by the coordinate planes, the cylinder $x^2+y^2=4$ and $z+y=3$. 5

2. a) i) Find the Laplace Transform of: $\frac{sin^2 t}{t}$ 8

ii) Find the inverse Laplace Transform of: $\frac{w}{s^2(s^2+w^2)}$

b) Solve the initial value problem: $y''+2y'$-3y=6e^{-2t} where, y(0)=2, $y'(0)$=-14 using Laplace Transform. 7

3. a) Solve $y''+y=0$ by using power series method. 7

b) Derive Rodrigue's formula for Legendre polynomials. 8

OR

Define Bessel's differential equation of order n. Find its complete solution.

4. a) Find the directional derivatives of $f = xy^2 + yz^3$ at P(2, −1, 1) along the direction of the normal to the surface x logz − y^2 + 4 = 0 at (−1, 2, 1). 5

b) If $\phi = \log(x^2+y^2+z^2)$, find div(grad ϕ) and curl(grad ϕ). 5

c) Show that the value under integral sign is exact and then evaluate integral, 5

$$\int_{(0,2,3)}^{(1,1,1)} [yz \sinh (xz)\, dx + \cosh (xz)\, dy + xy \sinh (xz)\, dz].$$

5. a) Let S be the hemi-sphere given by $z = \sqrt{9 - x^2 - y^2}$ and $\vec{F} = (3x, 3y, z)$. Find the flux of $\vec{F}$ through S. 7

b) State Gauss divergence theorem. Evaluate $\iint_S \vec{F}.\vec{n}\, dA$ where $\vec{F} = (y^3, x^3, z^3)$ and S is the cylinder $x^2 + 4y^2 = 1$ $0 < z < 5,\ x > 0, y > 0$ by using it. 8

OR

State Stoke's theorem. Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = (y^3, 0, x^3)$ and C is the boundary of the triangle with vertices are given by the vectors (1, 0, 0), (0, 1, 0), (0, 0, 1).

6. a) Find the fourier series of $f(x) = x + |x|$ for $-\pi < x < \pi$. 7

b) Find the fourier cosine series of $f(x) = x \sin x$, $0 < x < \pi$ then show that $\frac{\pi-2}{4} = \frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \frac{1}{7.9} + \cdots$. 8

7. Attempt **any two** questions. 2×5

a) Solve: $u_t + 2uu_x = 0, u(x, 0) = e^{-x}$.

b) Derive the one-dimensional traffic flow model using conservation law.

c) Find the angle between the surfaces $xy^2z - 3x - z^2 = 0$ and $3x^2 + 2z = 1 + y^2$ at (1, -2, 1)